



Department of Scientific Computing, Modeling and Simulation

SC: 504 - Computational Lab - I

Test - 4

Sem - I, M.Sc.(Scientific Computing)

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Time: 10:15 AM to 11:15 AM

Date: September 30, 2025

Max mark: 20

Note:

1. Write the time and space complexity for each program. This part is optional and has no marks.

1. Attempt ALL

(a) Write a C program that:

(4)

1. Reads n integers into a dynamically allocated array.
2. Uses a **for** loop with **if--else if--else** conditions to count how many numbers are:
 - Positive
 - Negative
 - Zero
3. Uses **nested loops** to print all pairs of elements in the array where the sum of the pair is positive.

Example Input:

```
5
3 -1 0 4 -2
```

Example Output:

```
Positive count: 2
Negative count: 2
Zero count: 1
Pairs with positive sum:
(3, -1)
(3, 0)
(3, 4)
(3, -2)
(-1, 4)
(0, 4)
(4, -2)
```

(b) Write a C program that prints the following pattern using a **while** loop:

(5)

Accept n from the user. For example $n = 5$, the output should be:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

(c) Write a C program to

1. Reads n integers into a dynamically allocated array.

(1)

2. Perform the following on the array :

(1)

i. Replace each even number with 0.

(4)

ii. Replace each odd numbers with 1.

iii. Sort the modified array in non-decreasing order.

Example :

`arr = [4, 3, 2, 1]`

Output: `[0, 0, 1, 1]`

- Replace the even numbers (4 and 2) with 0 and the odd numbers (3 and 1) with 1. Now, `arr = [0, 1, 0, 1]`.

- After sorting `arr` in non-descending order, `arr = [0, 0, 1, 1]`.

(d) Write a C program to read n integers into a dynamically allocated array. Then count the number of good pairs in this array. A pair (i, j) is called good if `arr[i] == arr[j]` and $i < j$.

(5)

Example :

Input: `arr = [1, 2, 3, 1, 1, 3]`

Output: 4

Explanation: There are 4 good pairs :
(0,3), (0,4), (3,4), (2,5) 0-indexed.

>>> Practice >> Consistency >> Perfection.